

Lab2 Guide Study the characteristics of basic electronic components and Circuit simulation with Multisim

Objectives

To learn some useful features of the Multisim simulation software and to highlight some differences between the computations done in class and the results of Multisim simulations and benchtop experiments.

To reinforce the theory of diode and measure it in the lab.

To reinforce the theory and certain operation of capacitor.

Familiarize yourself with Signal generator, DMM and oscilloscope.

Introduction to CAD Tools

Practical circuit design occurs in three stages:

1. Design of an appropriate circuit diagram and calculation of all component values to meet the specifications.
2. Verification of the design with circuit simulation.
3. Building and testing the circuit in the laboratory.

The building phase usually takes considerable time and resources. Tracking down design errors in the laboratory can take hours. The turn-around time for fabricating and stuffing printed circuit boards is days to weeks, and fabricating custom integrated circuit takes several months, not to speak of the high cost.

Circuit simulation allows verifying circuit performance before committing to fabrication. Errors found at this stage are much less time consuming and costly to fix. Although some circuits designed are relatively simple and could perhaps be successfully designed without simulation, every experienced engineer can tell you about disasters where a circuit that was supposedly too simple to simulate did not work properly malfunctioned and held up a larger project. Many successful electronics companies have policies allowing circuits to be built only after they have been successfully simulated.

This lab will introduce you the simple use of one circuit simulation tool, MultiSim. And we will provide a brief introduction about another circuit simulation tool, PSpice.

PSpice

PSpice SPICE simulation technology combines industry-leading, native analog and mixed-signal engines to deliver a complete circuit simulation and verification solution. It meets the changing simulation needs of designers as they progress through the design cycle, from circuit exploration to design development and verification. Designed for use in conjunction with PSpice A/D, PSpice Advanced Analysis helps designers improve yield and reliability. If you want to download the PSpice by yourself, the website is below.

<http://www.orcad.com/resources/orcad-downloads>

MultiSim

Multisim is a schematic capture and electronic circuit simulator supplied by National Instruments (NI). It runs as an application on the PC. You may think of Multisim as a virtual laboratory. You will mainly use this simulator in pre-lab and post-lab activities. The results of your simulations represent the ideal circuit response. In your lab reports, you will use those results to benchmark the experimental results obtained from the actual physical circuit. It is quite **important** that before running a simulation, **you should always have a general understanding of how your circuit works.**

The MultiSim can be downloaded at the website below:

<http://www.ni.com/multisim/student-edition/zhs/>

The basic Circuit Simulation Techniques in MultiSim should be familiar with are:

- MultiSim Environment
- Schematic Capture of an Example Circuit
- Simulation and Results Display

Diodes Overview

Typical diode packages in same alignment as diode symbol where thin bar depicts the cathode as shown in Figure 1(a). For Light-Emitting Diode (LED), the longer lead usually indicates the anode as shown in Figure 1(b).

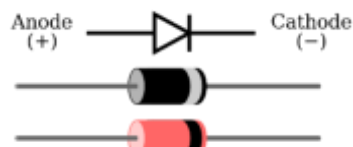


Figure 1(a). Orientations of typical diode

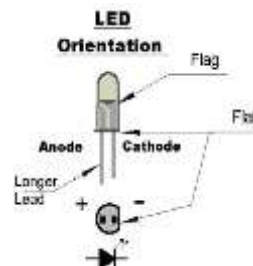


Figure 1(b). LED.

An ideal diode has an infinite resistance when the voltage across it is less than its “threshold voltage” (or $V_{\text{threshold}}$) and zero resistance when the voltage is greater than the threshold. The threshold voltage is just a characteristic of each individual diode i.e. every 1N4148 diode should have the same threshold voltage (around 0.6 volts) whereas an LED may have a different threshold voltage. This threshold voltage concept comes from the fact that a diode is just a p-n junction. Don’t feel bad if you haven’t studied p-n junctions before, it is not required for this lab. The “I-V” graph for an ideal diode looks like that in Figure 2.

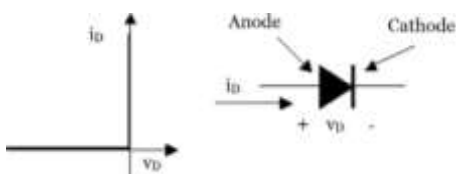


Figure 2(a). Ideal Diode I-V Curve

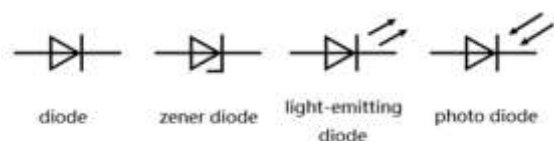


Figure 2(b) Symbols of different types of diodes

In Figure 2, the threshold voltage (i.e., the voltage when the slope of the line changes from 0 to ∞) is at 0. This will not be the case for the real diodes we use in lab. For the diodes we will use in this lab, all threshold voltages will be positive. We can find that the behavior of diodes is actually somewhat like a switch.

Test diodes with Digital Multimeter

Digital multimeters can test diodes using one of the two following methods: diode test mode (usually the better approach) and resistance mode (typically used only if a multimeter is not equipped with a Diode Test mode.) A diode is best tested by measuring the voltage drop across the diode when it is forward-biased.

A multimeter's Diode Test mode produces a small voltage between the test leads. The multimeter then displays the voltage drop when the test leads are connected across a diode that is forward-biased. The Diode Test procedure is conducted as follows:

1. Make certain all power to the circuit is OFF and no voltage exists at the diode. Voltage may be present in the circuit due to charged capacitors. If so, the capacitors need to be discharged. Set the multimeter to measure ac or dc voltage as required.
2. Turn the dial (rotary switch) to Diode Test mode ($\rightarrow +$). It may share a space on the dial with another function.
3. Connect the test leads to the diode. Record the measurement displayed.
4. Reverse the test leads. Record the measurement displayed.

A good forward-biased diode displays a voltage drop ranging from 0.5 to 0.8 volts for the most commonly used silicon diodes. The multimeter displays OL when a good diode is reverse-biased. The OL reading indicates the diode is functioning as an open switch.

Diode I-V characteristics

The I-V graph for a non-ideal diode is shown in Figure 3, along with an ideal approximation to accommodate the non-zero threshold voltage. The diode will be easier to understand if we compare the diode to another two terminal device we know (and love) - the resistor.

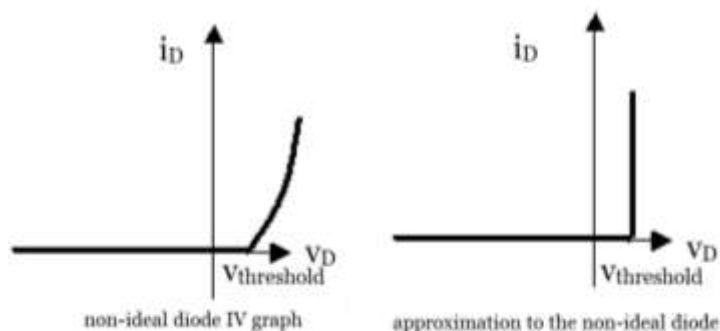


Figure 3. Non-Ideal Diode I-V Curve and an approximation to the non-ideal diode

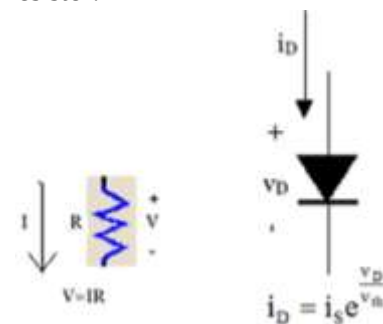


Figure 4. The resistor vs. the diode. V_{th} is the threshold voltage

From Figure 4, we see that both diodes and resistors are two terminal devices. However, their I-V characteristics are very different. An equation that models the I-V characteristic of a non-ideal diode is shown below. If v_D is greater than v_{th} , then the diode is said to be forward-biased or it is said to be in the forward-biased region. If not, the diode is said to be operating in reverse-bias. Also, in the equation:

$$i_D = I_S e^{\frac{v_D}{V_{th}}} \quad \text{if } v_D \geq v_{th}$$

$$i_D = 0 \quad \text{if } v_D \leq v_{th}$$

I_S is a constant called the reverse bias saturation current and is approximately equal to 1×10^{-11} A.

V_{th} is a constant called the thermal voltage (this is different from the threshold voltage) and is approximately equal to 26 mV at room temperature.

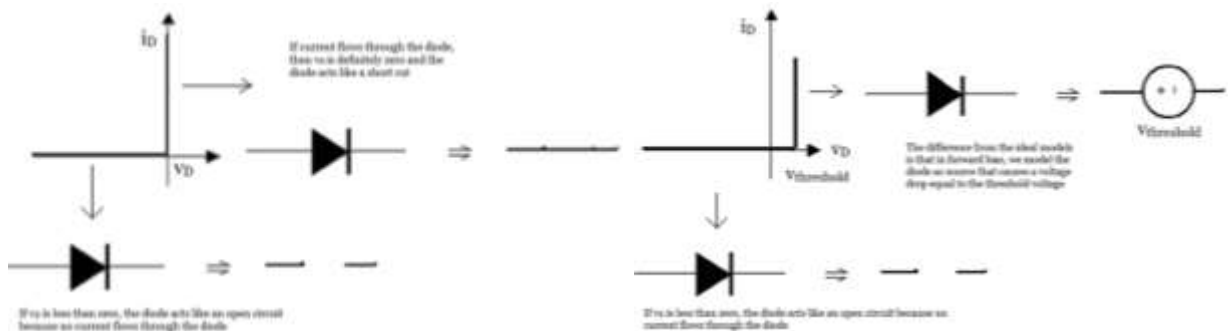


Figure 5. Ideal diode model with zero threshold voltage.

Figure 6. Ideal diode model with positive threshold voltage.

One more property of the diode --- looking at Figures 5 and 6, if you think about the diode symbol as an arrow --- you can infer that current can flow through the diode only in the direction of the arrow. Let us apply these two models and study the very practical diode circuit shown in Figure 7 --- the half-wave rectifier.

Half-Wave Rectifier

The half-wave rectifier is a circuit that allows only part of an input signal to pass. The circuit is simply the combination of a single diode in series with a resistor, where the resistor is acting as a load (see Figure 7 below).

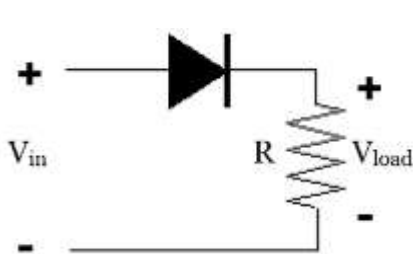


Figure 7. Half-Wave Rectifier Schematic

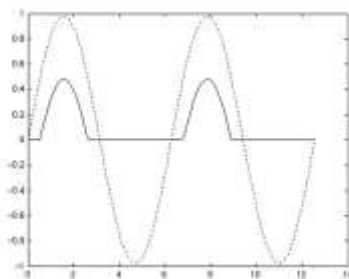


Figure 8. Half-Wave Rectifier, Voltage vs Time, V_{load} and V_{in} from Figure 7 are plotted. The dotted line is the input sinusoid (V_{in})

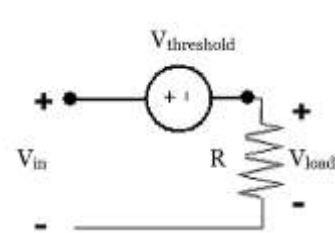


Figure 9. Half-Wave Rectifier with threshold voltage model when the diode is forward biased.

The output from the half-wave rectifier is shown in Figure 8. We can see that if the V_{in} is greater than zero (corresponding to a positive half-cycle on the sinusoid), the diode is forward biased. Using the threshold voltage model from Figure 6, we can redraw the circuit in Figure 7 as shown in Figure 9.

Hence, the effect of the diode is to drop a voltage of $v_{threshold}$ from the input. You can see this effect in Figure 8, the peak V_{load} voltage is less than V_{in} by $v_{threshold}$. When the diode is reverse-biased, that is when the V_{in} is the negative half-cycle of the sine wave, the diode is off and hence it is modeled as an open circuit. Thus, the current flowing through the circuit is zero and $V_{load} = 0$. This explains what happens during the negative half-cycles of the sinusoid in Figure 8.

Capacitor

A capacitor is an electrical device for storing charge. In general, capacitors are made from two or more plates of conducting material separated by a layer or layers of insulators. The capacitor can store energy to be returned to a circuit as needed. There are many different types of capacitors, each with a different dielectric material between the two electrodes and a different electrode shape. The main types of capacitors such as electrolytic, tantalum, mylar, ceramic, polystyrene and so on.

1. Polarized capacitors (generally large values, $\geq 1 \mu F$)

Electrolytic capacitors are polarized capacitors are large capacitors both in volume and capacitance. They look like a cylinder, as shown in Figure 1, it is easy to find the value of electrolytic capacitors because they are clearly printed with their capacitance and voltage rating. The capacitance value of the electrolytic capacitors in Figure1 is 1000 μF , and voltage rating are 16V and 25V, respectively.

As shown in Figure10, electrolytic capacitors are polarized, plus sign (for the longer leg) or a minus sign (for the shortest leg) indicating positive or negative terminal, respectively. Be careful when using these capacitors, if you put a voltage of the wrong polarity over electrolytic capacitor, the capacitor will be destroyed and may even explode! Capacitors also have voltage ratings, never put a larger voltage over the capacitor than what is specified on the capacitor. This is also the case for all other capacitor types.



Figure 10. electrolytic capacitors and symbols

2. Unpolarized capacitors (small values, up to 1 μF)



Figure 11. (a) Mica (b)Ceramic (c)polyester



(d)symbol



Figure12.

Reading Unpolarized capacitance values can be tricky and is generally more difficult than reading polarized capacitance values. The unit of a capacitance is the Farad, One Farad is a huge capacitor, both in value and physically. But as mention before, unpolarized capacitors are usually small values, as a result, capacitors are usually expressed in microfarads ($\times 10^{-6}\text{F}$) for large capacitors, nano farads ($\times 10^{-9}\text{F}$) and picofarads ($\times 10^{-12}\text{F}$) for small capacitors.

As shown in Figure.12, a number code is often used on small capacitors where printing is difficult:

the 1st number: the 1st digit;

the 2nd number: the 2nd digit;

the 3rd number: the number of zeros to give the capacitance in **pF**.

For example: 104 translates to $100000\text{pF} = 100\text{nF} = 0.1\mu\text{F}$ (not 104pF !)

Square Wave

The square shaped electrical waveforms are symmetrical in shape as each half of the cycle is identical, so the time that the pulse width is positive must be equal to the time that the pulse width is negative or zero. When square wave waveforms are used as “clock” signals in digital circuits the time of the positive pulse width is known as the “Duty Cycle” of the period. Then we can say that for a square wave waveform the positive or “ON” time is equal to the negative or “OFF” time so the duty cycle must be 50%, half of its period. As frequency is equal to the reciprocal of the period, $(1/T)$ we can define the frequency of a square wave waveform as:

$$f = \frac{1}{\text{ON time} + \text{OFF time}}$$

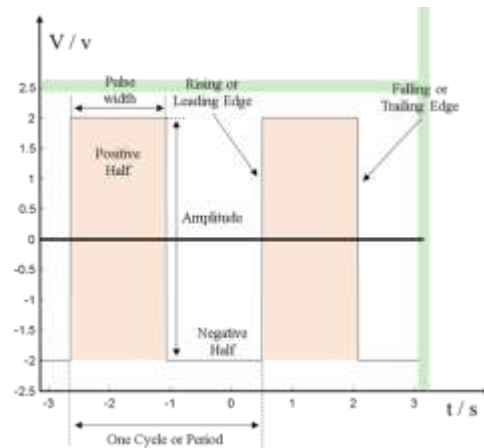


Figure 13. Square Wave

Rectangular Waveform

Rectangular Waveforms are similar to the square wave waveform above, the difference being that the two pulse widths of the waveform are of an unequal time period. Rectangular waveforms are therefore classed as “non-symmetrical” waveforms as shown below. The duty cycle is defined as the time a rectangular wave is at high value with respect to the period of the rectangular wave. It is usually indicated in %, e.g., if the period of the rectangular wave is 10 millisecond and the rectangular is in high position for 3 milliseconds, then the duty cycle is $(3/10) \times 100\% = 30\%$.